PYTHON PROGRAMMING

II B. TECH- I SEMESTER								
Course Code:	**Category**	**Hours / Week**			**Credits**	**Maximum Marks**		
A5IT03	**PCC**	**L**	**T**	**P**	**C**	**CIE**	**SEE**	**Total**
		3	1	-	4	30	70	100

COURSE OBJECTIVES

1. Understand the basics and function of Python Programming Language.
2. Understand the string operation and sequences used in Python Programming Languages.
3. Understand the data structures used in Python Programming Languages.
4. Know the classes and objects in Python Programming Language.
5. Use the reusability concepts in Python Programming Language.

COURSE OUTCOMES

1. Apply control structures and functions in Program
2. Analyze various String handling functions and data structures
3. Apply the files and object orientation concepts
4. Solve the problems by using Inheritance and polymorphism
5. Illustrate programs on various python libraries such asnumpy, pandas and matplotlib

SYLLABUS

UNIT - I	Introduction to Python Programming	CLASSES: 12

Introduction to Python Programming: Features of Python Language, Data Types, Arithmetic Operators, Comparison (Relational) Operators, Assignment Operators, Logical Operators, Bitwise Operators, Shift Operators, Ternary operator, Membership Operators, Identity Operators, Expressions and order of evaluations. Illustrative examples on all the above operators, Expressions, Control Statements, and Standard I/O Operations: input, print, 'sep','end'.

Functions and Modules:
Declaration and Definition Function Calling, Defining Functions, Recursive Functions, Modules, Packages in Python, Doc Strings, Built-in Functions.

UNIT – II	Strings and Regular Expressions	CLASSES: 13

Strings and Regular Expressions:
String Operations, Built-in String Methods and Functions, Comparing Strings, functions in Regular Expression.
Sequence: List, Tuples, Dictionaries.

UNIT - III	File Handling	CLASSES: 12

File Handling: Introduction, File Path, Types of Files, Opening and Closing Files, Reading and Writing Files, File Positions, Renaming and Deleting Files, Directory Methods.

Implementation of classes and objects in Python:

Classes and Objects, Methods and Self Argument, The __init__Method, Class Variables and Object Variables, The __del__Method, Public and Private Data Members, Private Methods,Built-in Functions to Check, Get, Set and Delete Class Attributes, Garbage Collection (Destroying Objects).

UNIT – IV	Implementation of Inheritance in Python	CLASSES: 11

Implementation of Inheritance in Python: Inheriting Classes in Python, Types of Inheritance, Composition/Containership, Abstract Classes and Interfaces, Meta class,Implementing Operator Overloading, Overriding Methods **Exception Handling in Python:** Introduction, Exception hierarchy, Handling Exception, Multiple except Blocks and Multiple Exceptions, Finally Block.		
UNIT - V	**Python NumPy**	**CLASSES: 10**
Python NumPy: Features of Numpy, NumPyndarray, Data Types, Functions of NumPy Array, Numpy Array Indexing, Mathematical Functions on Arrays in NumPy. **Python Pandas:** Pandas Features, Install Pandas, Dataset in Pandas, Series, DataFrames, Panel, Manipulating the Datasets, Describing a Dataset, group by Function, Filtering, Missing Values in Pandas, Concatenating Data Frames ***Matplotlib:*** *Formatting the style of plot, plotting with keyword strings, plotting with categorical variables, controlling line properties.*		

TEXTBOOKS

1. Core Python Programming, by R.NageswaraRao
2. ReemaThareja, Python Programming using Problem Solving Approach, First Edition,Oxford Higher Eduction.

REFERENCE BOOKS

1. Kenneth A.Lambert, Fundamentals of Python
2. Charles Dierach, Introduction to Computer Science using Python

SUGGESTED LINKS

1. https://www.programiz.com/python-programming
2. https://www.javatpoint.com/python-tutorial
3. https://www.geeksforgeeks.org/python-programming-language/